DEPARTMENT OF COMPUTER SCIENCE

Senior Capstone Project Proposal

Autonomous Room Scanner

A title is required even in draft form — a placeholder that names the problem or system (e.g., "A Mobile App for Campus Food-Pantry Scheduling") is far more useful to reviewers than "My Capstone Project."

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Team / Group Number (if applicable):

Faculty Advisor: Fred Agbo

Course / Section: 480W

Submission Date: **Sep 21, 2026**

How to use this template

This is a working document, not a final contract. Every section includes a short italicized prompt describing what a reviewer expects to see there — write your best current thinking underneath each one, even if it is still rough, incomplete, or likely to change. A proposal with honest gaps and open questions is more useful, and more gradable, than one that hides uncertainty behind vague language.

Target length: approximately 5 pages (excluding the title page and references), single-spaced, 11–12 pt font.

1. Problem Statement and Motivation

Describe, in plain language, the real problem or unmet need your project addresses and why it matters to actual users or stakeholders.

> ***Consider:*** *who experiences this problem today, how they currently cope without your solution, and what evidence (personal experience, articles, data, interviews) suggests the problem is real and worth solving.*

It is a cumbersome process to take 3D scans of rooms. Pictures must be taken manually from many angles and positions to create a usable scan. I aim to make the process more easy and enjoyable by designing a small robot to navigate through and scan a room, generating a high-fidelity scan with Gaussian Splatting. This project is not only motivated by the prospect of improving methods for 3D scanning, but also by my own interest in supporting computer vision tasks with robotics.

2. Project Overview

Summarize, in 3–5 sentences, what you plan to build or investigate, as if explaining it to a classmate outside the CS department.

> ***Consider:*** *naming the type of artifact (web app, mobile app, ML model, systems tool, research study, etc.) and the single core function it must perform.*

I plan to build a small wheeled robot powered by a single-board computer that is capable of navigating around a room and generating a 3D model of the space. The robot may be piloted by a user from a nearby laptop, or autonomously drive itself around the room, capturing images as it moves. The 3D model will be generated using a technique called Gaussian Splatting, and may be imported into third-party software such as Unity or Blender.

3. Goals and Objectives

List the specific, measurable objectives your finished project should achieve, distinguishing must-have goals from stretch goals.

> ***Consider:*** *phrasing each objective so a reader could later check yes/no whether it was met (e.g., "users can upload a file under 10MB and receive results in under 5 seconds").*

Must-have goals include:

- A custom-built vehicle equipped with a camera capable of being manually piloted around a room.
- Camera output feeds to an external device (such as a nearby laptop) that uses COLMAP and GaussianSplatting to generate a 3D model of the room. This file can be imported into a scene in Unity.
- The robot streams a lower resolution live video stream to the external device as it traverses the room.

Stretch goals include:

- Integrating a vertical-panning camera using a servo motor to capture more angles for a more accurate 3D model.
- A camera mast to capture more appropriate angles for 3D reconstruction.
- Multiple (2, potentially 3) cameras for a higher quality 3D model.
- Autonomously navigates a room utilizing a roomba-style traversal algorithm.

4. Scope and Boundaries

Clarify what is explicitly included in this project and what is explicitly excluded, to prevent scope creep later.

> ***In scope:*** *the features, platforms, or use cases you commit to delivering by the deadline.*
>
> ***Out of scope:*** *related ideas you are consciously setting aside for a future version or a different project.*

In Scope:

- Constructing a robot from scratch that can be piloted from an external device.
- Using camera output from the robot to create a 3D model of a room.
- Streaming a live video feed to the external device.

Out of Scope:

- Intelligently adjusting cameras' angle or positioning based on surroundings (e.g. detecting that the robot is under a table, and automatically panning the camera upwards to make sure the underside of the table is captured.)
- Incorporating advanced sensors such as LIDAR for better obstacle detection and navigation.
- Ability to intelligently avoid including humans in the generated 3D model if humans are present in the room during scanning.

5. Technical Approach and Methodology

Outline your current best guess at the technologies, architecture, algorithms, or methods you expect to use, and how you will verify the result works.

> ***Consider:*** *languages/frameworks under consideration, major components or modules, data sources, and how you will test or evaluate correctness and performance.*
>
> ***It's okay if:*** *this section is tentative — flag the specific technical decisions still open and what you'd need to learn or prototype to resolve them.*

Hardware

- **Robot car chassis:** I intend to acquire a car chassis from an online retailer. I am looking for a chassis that includes the frame, wheels, TT motors, and nothing more. There are chassis available that include more components, such as a motor driver, microcontroller, sensors, etc. however I am avoiding these as they abstract an important set of the skills I wish to gain from this project. I am interested in some that have omnidirectional Mecanum wheels, which I expect would simplify autonomous navigation.
- **Board:** From my research, it seems like a *Raspberry Pi 4B* would be a good fit. It is a single board computer running a distribution of Linux, and can be equipped with 1GB, 2GB, 3GB, 4GB, or 8GB of ram. All 3D modeling and point-cloud generation will be done on the external device, so I don't expect to need much ram on-device.
- **Camera:** I am looking to use a *Raspberry Pi Camera 4*. The camera works natively with a Raspberry Pi and is affordable for the image quality. It comes in Standard, Wide, and NoIR (No Infrared Filter) versions. I will

decide between the Standard and Wide versions, as the NoIR version is best suited for very dark conditions. I believe the wider FOV in the Wide camera (120 degrees vs 75 degrees) is better suited for my task of scanning a room, however before purchasing I first want to ensure that any barrel-distortion from the wide lens will not cause downstream problems in the 3D reconstruction pipeline.

- **Motor Driver:** To properly use the motors which drive the wheels, I need a motor driver. At the bare minimum, I need a motor driver that supports 2 motors moving forward and backwards (such as an *L298N Module*), however it would be best to use a motor driver that can support 4 motors moving forward and backwards (a 4-motor driver, such as the *Adafruit DC & Stepper Motor HAT*, is required if I want to implement omnidirectional driving with Mecanum wheels.) Most likely I will go with the *Adafruit DC & Stepper Motor HAT*.
- **Power Supply:** Both the Raspberry Pi 4B and the motor driver will need their own power supply. If the Raspberry PI 4B powers the motors directly, voltage spikes from the motors risk frying the board.

Software

- **Language:** I will use Python to program both the on-device logic and the 3D model pipeline. I am most comfortable with Python and there are many useful libraries both for working with hardware on a Raspberry Pi and for using COLMAP and GaussianSplatting, mentioned below.
- **Point-Cloud Generation:** I will use *COLMAP* via the *pycolmap* library to get point-cloud information from images captured by the robot. I will use the *hloc* library with *pycolmap* to GPU accelerate the process.
- **3D Model Construction:** I will use Gaussian Splatting, likely with either the *gsplat* or *Nerfstudio* libraries. Gaussian Splatting constructs high-fidelity 3D models quickly. Compared to alternatives like NeRF and Photogrammetry, Gaussian Splatting offers the best trade-off between model quality, compute needed, and run time.
- **Device Communication:** I am planning for the robot and the external device to communicate with each other using Websockets over WiFi. This approach should offer low latency, and allow the robot to be controlled from a laptop or from a mobile device.
- **Autonomous Navigation:** Since this is a stretch goal, more research must be done on this topic. At the moment, I think that it would be relatively simple to implement a primitive roomba-style algorithm that moves the robot semi-randomly around the room until it has achieved good coverage. However, this may not be robust enough to create an accurate and complete 3D model of a room, in which case I will seek out alternative algorithms.

6. Related Work and Existing Solutions

Briefly identify at least two or three existing tools, products, papers, or approaches that address a similar problem, and state how your project will differ or improve on them.

> ***Consider:*** *a short table or bullet list comparing each alternative's strengths and limitations against your intended approach.*

The current methods for generating a 3D model of a room often use either a smartphone or extremely expensive professional equipment, both of which have significant tradeoffs. The smartphone solutions most often require that the user pan around and record the space, creating a 3D scan from the output. These smartphone-based techniques prioritize quick feedback over generating a high-quality model. On the other hand, professional 3D-scanning equipment can create extremely high-fidelity 3D scans, but is prohibitively expensive for most use cases. Both methods require photo or video being manually taken of the scene. My solution, however, takes the manual

photo-taking out of the equation- and while it isn't expected to match the quality of professional-grade equipment, the use of Gaussian Splatting is expected to produce much higher quality 3D models as compared to the instant-feedback smartphone apps.

7. Timeline and Milestones

Break the semester into 4–6 milestones with target dates and a one-line deliverable for each, working backward from the final deadline.

Target Date	Milestone / Deliverable	Status
Oct 5, 2026	Get an MVP of COLMAP + Gaussian Splatting working with manually-taken images.	Not Started
Oct 12, 2026	Purchase all necessary hardware.	Not Started
Nov 2, 2026	Assemble the robot until it is driveable.	Not Started
Nov 16, 2026	Implement Remote-Control functionality.	Not Started
Nov 30, 2026	Implement Image Capturing and Uploading.	Not Started

General timeline of the whole project:

- setup pipeline MVP with COLMAP and Gaussian Splatting. Use manually taken images to test it, and get GPU acceleration working with ROCm
- purchase all hardware
- assemble the robot to the point it is able to move
- implement remote control functionality
- make the robot able to take pictures at set intervals while moving
- create a protocol to send the images from the robot to the external device
- clean everything up and tackle a reach goal (autonomous navigation and improved cameras)

8. Anticipated Challenges and Risks

Name the two or three biggest things that could go wrong (technical, scheduling, data access, scope) and your current plan for handling each.

> ***Consider:*** *a simple risk / mitigation pairing rather than a general worry — e.g., "Risk: needed dataset may be restricted. Mitigation: identify a public substitute by week 3."*

- **Extremely Long Runtime**
    - **Risk:** COLMAP and Gaussian Splatting rely on GPU acceleration to keep runtime manageable. Since COLMAP uses SIFT under the hood, which compares every image to every other image, it has a quadratic time complexity, which could lead to exploding compute times if not properly managed. The GPU acceleration is designed to natively use CUDA, which is exclusive to NVIDIA GPUs. However my laptop has an AMD GPU (Radeon RX 6800S). AMD has an open-source tool called ROCm which is their alternative to CUDA, but it is reportedly finicky to work with.
    - **Mitigation:** To manage the time complexity, I will do many tests to strike a balance between high model quality and acceptable compute time, taking only as many images as necessary to create a high-fidelity image. I will use ROCm to GPU accelerate my 3D model generation.
- **Poor Model Generation Due to Low Camera**
    - **Risk:** The 3D model may not be as high quality as is desired due to the fact that all images will be taken from an angle low to the ground. This may result in objects like tabletops or shelves not being properly captured in the 3D model.
    - **Mitigation:** I will try adding a mast for the camera to be attached that sits roughly 3-4 feet above the robot. If this results in accurate reconstruction of taller objects, but poorer reconstruction of shorter objects, I will use 2 cameras, one placed on the mast and one placed on the robot closer to ground level. I can also try adding a servo motor to pan the camera up and down.

9. Expected Outcomes and Deliverables

List the concrete artifacts you will hand in at the end of the semester, such as source code, a deployed demo, a final report, a poster, or a dataset.

By the end of the project, I intend to produce the following:

- A custom-built robot capable of navigating around and taking pictures of a room.
- A pipeline using COLMAP and Gaussian Splatting to generate a 3D model of a room using the images captured by the robot.
- A public GitHub repository showcasing the source code for the project.
- A final report documenting the project.

10. Resources and Support Needed

Note any hardware, software licenses, datasets, cloud credits, faculty expertise, or external partners this project will require.

Hardware requirements are outlined in greater detail in **Technical Approach and Methodology** covering justification and alternatives, however below is a comprehensive list of what I expect to need for the project.

- Car chassis with 4 wheels and 4 TT Motors
- Raspberry Pi 4 Model B
- Motor driver
- 2x Power supply
- Raspberry Pi camera
- Small storage drive
- Soldering iron and related supplies

All software I expect to use is open source, so I do not expect to need support in attaining licenses. I intend to contact two professors for guidance regarding this project. I will continue to consult Professor Rachel Brown for computer vision support, especially given that her PhD dissertation is in a relevant domain of 3D scanning and reconstruction. I will also consult Professor Calvin Deutschbein for support with hardware-level programming.

11. References

Cite any articles, tools, APIs, or prior work referenced above, in a consistent citation style (e.g., IEEE or ACM).

Bernhard Kerbl, Georgios Kopanas, Thomas Leimkuehler, and George Drettakis. 2023. 3d Gaussian splatting for real-time radiance field rendering. ACM Transactions on Graphics 42, 4 (July 2023), 1–14. DOI:http://dx.doi.org/10.1145/3592433

Colmap. Colmap/colmap: COLMAP - structure-from-motion and Multi-View Stereo. Retrieved September 21, 2026 from https://github.com/colmap/colmap